through (d)(4) should be increased,[1485] the Commission is modifying its initial burden and cost estimates for new entrants, as discussed below.

(iii)Adopted Estimates – Initial Burden and Costs

The Commission is increasing its estimated initial costs and associated burden hours for new entrants that choose to become competing consolidators to build systems to comply with Rules 614(d)(1) through (d)(4). The Commission preliminarily estimated initial burden hour and external cost estimates for new entrants that are higher than those estimated for the potential entities, other than SROs, that may choose to become competing consolidators. Because new entrants would be wholly new to the business of consolidating market data, the Commission continues to believe that new entrants would incur substantially higher burden hours and external costs to build new systems to comply with Rules 614(d)(1) through (d)(4) than potential competing consolidators that already collect and aggregate market data.[1486] Additionally, the Commission initially believed that competing consolidators would build aggregation systems in a single data center; however, the Commission now believes that competing consolidators may build systems for aggregating data in more than one data center. The Commission is increasing its estimated initial burden hours for new entrants to build systems to comply with Rules 614(d)(1) through (d)(4). The Commission also believes new entrants would likely incur external costs greater than the Commission's estimate to buy new technology (for example, hardware and network infrastructure).

1485 IDS Letter I at 13, n.38.

1486 The Commission's assumption is supported by a commenter, which stated, "The incumbent SIPs, the Securities Industry Automation Corporation ('SIAC') and Nasdaq UTP, will have a significant competitive advantage over new entrants should they chose [sic] to transition to Competing Consolidators. For example, the incumbent SIPs will benefit from utilizing the existing infrastructure, which was funded by industry participants, to transform to a Competing Consolidator." MIAX Letter p. 2–3.

As it did for its market data aggregation firm estimates, the Commission is increasing its estimated burden hours for Sr. Programmers and Sr. Systems Analysts by three times for new entrants, as well as its estimated external costs to be incurred by new entrants to purchase new technology to upgrade their systems to comply with Rules 614(d)(1) through (d)(4). The Commission estimates that each new entrant would incur 8,800 initial burden hours to build systems to comply with Rules 614(d)(1) through (d)(4),[1487] and initial external costs of $2,475,000 to purchase the necessary technology to build such systems,[1488] $194,000 to establish co-location connectivity to the exchange data centers,[1489] and $14,000 to purchase market data from the exchanges,[1490] for a total external cost to each new entrant of $2,683,000.[1491] The

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1487 The Commission estimated the monetized initial burden for this requirement to be $2,788,600. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 4,200 hours) + (Sr. Systems Analyst at $285/hour for 3,600 hours) + (Compliance Manager at $310/hour for 400 hours) + (Director of Compliance at $489/hour for 200 hours) + (Compliance Attorney at $366/hour for 400 hours)] = 8,800 initial burden hours to build systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1488 This estimate was originally based on discussions with a market participant and the Commission's understanding of hardware costs. The Commission has increased this estimated cost by three times because the Commission believes competing consolidators may potentially build aggregation systems in three data centers.

1489 This estimate is based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

1490 As it did in the Proposing Release, the Commission is using the monthly market data access and redistribution fees charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1491 The Commission estimated that each new entrant would incur the following initial external costs: [($2,475,000 to build systems to comply with Rules 614(d)(1) through

Commission estimates that the total initial burden hours for two new entrants would be 17,600 burden hours,[1492] and that total initial external costs would be $5,366,000 for two new entrants to build systems to comply with Rules 614(d)(1) through (d)(4).[1493]

(d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] = $2,683,000.

1492 The Commission estimated the monetized initial burden for this requirement to be $5,577,200. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 4,200 hours) + (Sr. Systems Analyst at $285/hour for 3,600 hours) + (Compliance Manager at $310/hour for 400 hours) + (Director of Compliance at $489/hour for 200 hours) + (Compliance Attorney at $366/hour for 200 hours)] x [(2 new entrants)] = 17,600 initial burden hours across the new entrants.

1493 The Commission estimated that the new entrants would incur the following initial external costs: [($2,475,000 to build systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] x [(2 new entrants)] = $5,366,000.

(f) Initial Burden and Costs for SROs[1494]

(i) Proposed Estimates – Initial Burden and Costs

In the Proposing Release, the Commission estimated that each SRO would incur 3,600 initial burden hours[1495] and $825,000 in external costs[1496] to build systems to comply with Rules 614(d)(1) through (d)(4). Additionally, the Commission estimated that an SRO would incur initial external costs of $14,000 to purchase market data from the SROs,[1497] and an additional initial external cost of $194,000 to co-locate itself at four exchange data centers,[1498] for a total

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1494 See supra note 1394.

1495 Based on discussions with a market participant, the Commission reached the following estimates for a market data aggregation firm: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 6 months (900 burden hours) to upgrade existing systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead. As it did for its new entrant estimates, the Commission increased this initial burden hour estimate to apply to the SROs. Therefore, the Commission estimated that each SRO will incur 3,600 initial burden hours to build systems to comply with Rules 614(d)(1) through (d)(4) (or $1,175,000, as monetized).

1496 As it did for its new entrant estimates, the Commission increased its initial external cost estimates for market data aggregation firms to apply to the SROs. Therefore, the Commission estimated that each SRO will incur $825,000 in initial external costs to build systems to comply with Rules 614(d)(1) through (d)(4).

1497 The Commission used the monthly market data access and redistribution fees currently charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1498 This estimate was based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

initial external cost of $1,033,000 per SRO,[1499] and an aggregate estimate of 14,400 initial burden hours[1500] and $4,132,000 in initial external costs.[1501]

(ii) Comments/Responses on Initial Burden and Costs

In response to the commenter that believed that the estimated costs incurred by potential competing consolidators to build or upgrade systems to comply with proposed Rules 614(d)(1) through (d)(4) should be increased,[1502] the Commission is modifying its initial burden and cost estimates for SROs, as discussed below.

(iii)Adopted Estimates – Initial Burden and Costs

The Commission is increasing its estimated initial costs and associated burden hours for SROs that choose to become competing consolidators to build systems to comply with Rules 614(d)(1) through (d)(4).[1503] The Commission initially believed and continues to believe that

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1499 The Commission estimated that each SRO would incur the following initial external costs: [($825,000 to build systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers)] = $1,033,000.

1500 Based on discussions with a market participant, the Commission reached the following estimates for a market data aggregation firm: [(Sr. Programmer at $332/hour for 350 hours) + (Sr. Systems Analyst at $285/hour for 300 hours) + (Compliance Manager at $310/hour for 100 hours) + (Director of Compliance at $489/hour for 50 hours) + (Compliance Attorney at $366/hour for 100 hours)] = 900 initial burden hours. As it did for its new entrant estimates, the Commission increased the per market data aggregation firm initial burden hour estimate to apply to the SROs. Therefore, the Commission estimated that each SRO would incur 3,600 initial burden hours to upgrade its existing systems to comply with Rules 614(d)(1) through (d)(4) (or $1,175,000, as monetized). [(3,600 burden hours) x (4 SROs)] = 14,400 hours (or $4,700,000 as monetized).

1501 The Commission estimated that each SRO would incur the following initial external costs: [($825,000 to build systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to co-locate within four exchange data centers)] = $1,033,000. [($1,033,000 in initial external costs) x (4 SROs)] = $4,132,000.

1502 IDS Letter I at 13.

1503 See supra note 1394.

these entities would have to build new systems to comply with Rules 614(d)(1) through (d)(4) and thus would incur initial burden hours that are similar to new entrants. Because SROs that do not operate exclusive SIPs would be wholly new to the business of consolidating market data, these entities would likely incur substantially higher burden hours and external costs to build new systems to comply with Rules 614(d)(1) through (d)(4) than potential competing consolidators that already collect and aggregate market data. Additionally, the Commission initially believed that competing consolidators would build aggregation systems in a single data center; however, the Commission now believes that competing consolidators may build systems for aggregating data in more than one data center. The Commission is increasing its estimated initial burden hours for SROs to build systems to comply with Rules 614(d)(1) through (d)(4). The Commission also believes that SROs would likely incur external costs greater than the Commission's estimate to buy new technology (for example, hardware and network infrastructure). The Commission is also revising its total initial burden hour and external cost estimates across these entities to reflect a reduction in the number of such competing consolidators.

As it did for its market data aggregation firm estimates, the Commission is increasing its estimated burden hours for Sr. Programmers and Sr. Systems Analysts by three times for SROs, as well as its estimated external costs to be incurred by SROs to purchase new technology to build systems to comply with Rules 614(d)(1) through (d)(4). The Commission estimates that each SRO would incur 8,800 initial burden hours to build systems to comply with Rules

614(d)(1) through (d)(4),[1504] and initial external costs of $2,475,000 to purchase the necessary technology to build such systems,[1505] $194,000 to establish co-location connectivity to the exchange data centers,[1506] and $14,000 to purchase market data from the exchanges,[1507] for a total external cost to each SRO of $2,683,000.[1508] The Commission estimates that the total initial burden hours for one SRO would be 8,800 burden hours[1509] and that total initial external costs

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1504 The Commission estimates the monetized initial burden for this requirement to be $2,788,600. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 4,200 hours) + (Sr. Systems Analyst at $285/hour for 3,600 hours) + (Compliance Manager at $310/hour for 400 hours) + (Director of Compliance at $489/hour for 200 hours) + (Compliance Attorney at $366/hour for 400 hours)] = 8,800 initial burden hours to build systems to comply with Rules 614(d)(1) through (d)(4). The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead.

1505 This estimate was originally based on discussions with a market participant and the Commission's understanding of hardware costs. The Commission has increased this estimated cost by three times because the Commission believes competing consolidators may potentially build aggregation systems in three data centers.

1506 This estimate is based on an estimated $48,500 in initial co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers. See NYSE Price List 2020, supra note 1434.

1507 As it did in the Proposing Release, the Commission is using the monthly market data access and redistribution fees charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000).

1508 The Commission estimates that each SRO would incur the following initial external costs: [($2,475,000 to build systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] = $2,683,000.

1509 The Commission estimates the total monetized initial burden for this requirement to be $2,788,600. These estimates were initially based on discussions with a market participant, modified as discussed above: [(Sr. Programmer at $332/hour for 4,200 hours) + (Sr. Systems Analyst at $285/hour for 3,600 hours) + (Compliance Manager at $310/hour for 400 hours) + (Director of Compliance at $489/hour for 200 hours) +

would be $2,683,000 for one SRO to build systems to comply with Rules 614(d)(1) through (d)(4).[1510]

(g) Ongoing Burden and Costs for Competing Consolidators

(i) Proposed Estimates – Ongoing Burden and Costs

In the Proposing Release, the Commission estimated that once a competing consolidator's system had been built, all types of entities that could become a competing consolidators (i.e., existing market data aggregation firms, broker-dealers that aggregate market data, exclusive SIPs, new entrants, and SROs) would incur annual ongoing burden hours and external costs to operate and maintain their systems to comply with Rules 614(d)(1) through (d)(4) and that the annual ongoing burdens would be similar for all types of competing consolidators because such systems would likely be similar in nature. Therefore, the Commission estimated the same annual ongoing burden hours and external costs for the five types of entities that the Commission anticipated may choose to become competing consolidators.

Competing consolidators would incur annual ongoing burden hours and external costs to operate and maintain their modified systems to comply with Rules 614(d)(1) through (d)(4). Specifically, the Commission estimated that each entity would incur 540 annual ongoing burden

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(Compliance Attorney at $366/hour for 400 hours)] x [(1 SRO)] = 8,800 total initial burden hours.

1510 The Commission estimates that SROs would incur the following total initial external costs: [($2,475,000 to build systems to comply with Rules 614(d)(1) through (d)(4)) + ($14,000 to purchase market data) + ($194,000 to establish co-location connectivity within four exchange data centers)] x [(1 SRO)] = $2,683,000.

hours[1511] and $123,725 in annual ongoing external costs[1512] to operate and maintain its systems to comply with Rules 614(d)(1) through (d)(4).

Further, the Commission estimated that each competing consolidator would incur annual ongoing external costs of $168,000 to purchase market data from the SROs,[1513] and an additional annual ongoing external cost of $4,602,720 to co-locate itself at four exchange data centers,[1514] for a total annual ongoing external cost of $4,894,445 per entity.[1515] In the Proposing Release,

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1511 The Commission estimated that once a competing consolidator's infrastructure was in place, the burden of operating and maintaining the infrastructure would be less than the burdens associated with establishing the infrastructure. The Commission estimated the monetized initial burden for this requirement to be $176,250. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Programmer at $332 for 210 hours) + (Sr. Systems Analyst at $285 for 180 hours) + (Compliance Manager at $310 for 60 hours) + (Director of Compliance at $489 for 30 hours) + (Compliance Attorney at $366 for 60 hours)] = 540 burden hours per entity and $176,250.

1512 This estimate was based on the initial external cost estimate for a market data aggregation firm to modify its systems to comply with Rules 614(d)(1) through (d)(4), but reduced because the Commission estimated that once a competing consolidator's infrastructure was in place, the burden of operating and maintaining the infrastructure would be less than the burdens associated with establishing the infrastructure.

1513 The Commission used the monthly market data access and redistribution fees currently charged by the CTA/CQ SIP and Nasdaq UTP SIP as the basis of this estimate ($14,000), multiplied by 12 months.

1514 This estimate was based on an estimated $95,890 in monthly co-location fees as calculated from NYSE Price List 2020, multiplied by four exchange data centers over 12 months. The Commission estimated that the market data aggregators would already be co-located at the four exchange data centers, which may lower this estimate for this category of respondent. See NYSE Price List 2020, supra note 1434.

1515 $4,894,445 = [($123,725 to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4)) + ($168,000 in monthly market data fees over 12 months) + ($4,602,720 to co-locate within four exchange data centers over 12 months)].

the Commission estimated that there would be two entities per category of potential competing consolidators for existing market data aggregators, broker-dealers that currently aggregate market data, exclusive SIPs, and new entrants, and that for each of these categories, the aggregate estimates would amount to estimate of 1,080 annual ongoing burden hours[1516] and $9,797,530 in annual ongoing external costs.[1517] In addition, the Commission estimated that there would be four SROs that would become a competing consolidator and that the SROs would

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1516 The Commission estimated the monetized annual ongoing burden for this requirement to be $352,500. The Commission derived this estimate based on per hour figures from SIFMA's Management & Professional Earnings in the Securities Industry 2013, modified by Commission staff to account for an 1,800-hour work-year and inflation, and multiplied by 5.35 to account for bonuses, firm size, employee benefits, and overhead: [(Sr. Programmer at $332 for 210 hours) + (Sr. Systems Analyst at $285 for 180 hours) + (Compliance Manager at $310 for 60 hours) + (Director of Compliance at $489 for 30 hours) + (Compliance Attorney at $366 for 60 hours)] x [(2 market data aggregation firms/broker-dealers that currently aggregate market data/exclusive SIPs/new entrants)] = 1,080 annual ongoing burden hours and $352,500.

1517 The Commission estimated that the market data aggregation firms/broker-dealers that currently aggregate market data for their own usage/exclusive SIPs/new entrants would incur the following aggregate annual ongoing external costs: [($123,725 to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4)) + ($168,000 in monthly market data fees over 12 months) + ($4,602,720 to co-locate within four exchange data centers over 12 months)] x [(2 entities)] = $9,788,890.